

Lab 1 - Introduction to Natural Language Processing (NLP)

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Section: 9MA1

```
In [1]: # Import pandas
import pandas as pd
```

```
In [2]: # Load the IMDB Dataset
df = pd.read_csv("IMDB Dataset.csv")
```

```
In [3]: df.head()
```

```
Out[3]:
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

```
In [4]: df.shape
```

```
Out[4]: (50000, 2)
```

```
In [5]: df.columns
```

```
Out[5]: Index(['review', 'sentiment'], dtype='object')
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   review      50000 non-null  object
1   sentiment   50000 non-null  object
dtypes: object(2)
memory usage: 781.4+ KB
```

```
In [7]: df.describe(include="all")
```

```
Out[7]:
```

	review	sentiment
count	50000	50000
unique	49582	2
top	Loved today's show!!! It was a variety and not...	positive
freq	5	25000

```
In [8]: df.isnull().sum()
```

```
Out[8]: review      0
sentiment    0
dtype: int64
```

```
In [9]: df["sentiment"].value_counts()
```

```
Out[9]: sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

```
In [10]: print(df["review"].iloc[0])
```

One of the other reviewers has mentioned that after watching just 1 Oz episode you'll be hooked. They are right, as this is exactly what happened with me.
The first thing that struck me about Oz was its brutality and unflinching scenes of violence, which set in right from the word GO. Trust me, this is not a show for the faint hearted or timid. This show pulls no punches with regards to drugs, sex or violence. Its is hardcore, in the classic use of the word.
It is called OZ as that is the nickname given to the Oswald Maximum Security State Penitentiary. It focuses mainly on Emerald City, an experimental section of the prison where all the cells have glass fronts and face inwards, so privacy is not high on the agenda. Emerald City is home to many...Aryans, Muslims, gangstas, Latinos, Christians, Italians, Irish and more....so scuffles, death stares, dodgy dealings and shady agreements are never far away.
I would say the main appeal of the show is due to the fact that it goes where other shows wouldn't dare. Forget pretty pictures painted for mainstream audiences, forget charm, forget romance... OZ doesn't mess around. The first episode I ever saw struck me as so nasty it was surreal, I couldn't say I was ready for it, but as I watched more, I developed a taste for Oz, and got accustomed to the high levels of graphic violence. Not just violence, but injustice (crooked guards who'll be sold out for a nickel, inmates who'll kill on order and get away with it, well mannered, middle class inmates being turned into prison bitches due to their lack of street skills or prison experience) Watching Oz, you may become comfortable with what is uncomfortable viewing....thats if you can get in touch with your darker side.

```
In [11]: print(df["sentiment"].iloc[0])
```

positive

```
In [12]: import pandas as pd

# Load dataset
df = pd.read_csv("IMDB Dataset.csv")

# Display first 5 rows
print("First 5 rows:")
display(df.head())
```

```

# Dataset shape
print("Dataset Shape:", df.shape)

# Column names
print("\nColumns:")
print(df.columns)

# Dataset information
print("\nDataset Information:")
df.info()

# Missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Sentiment distribution
print("\nSentiment Distribution:")
print(df["sentiment"].value_counts())

# Example review
print("\nExample Review:")
print(df["review"].iloc[0])

print("\nSentiment:")
print(df["sentiment"].iloc[0])

```

First 5 rows:

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

Dataset Shape: (50000, 2)

Columns:

Index(['review', 'sentiment'], dtype='object')

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 50000 entries, 0 to 49999

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	review	50000 non-null	object
1	sentiment	50000 non-null	object

dtypes: object(2)

memory usage: 781.4+ KB

Missing Values:

review 0

sentiment 0

dtype: int64

Sentiment Distribution:

sentiment

positive 25000

negative 25000

Name: count, dtype: int64

Example Review:

One of the other reviewers has mentioned that after watching just 1 Oz episode you'll be hooked. They are right, as this is exactly what happened with me.

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Sentiment:

positive